Name
Date
Course

Evaluating Algebraic Expressions (Part II)

① $x^2 - 2x + 1,\ x = -3$

$(-3)^2 - 2(-3) + 1$

$9 + 6 + 1$

$\boxed{16}$

② $-y^2 + \frac{y+2}{3} - 1,\ y = -1$

$-(-1)^2 + \frac{-1+2}{3} - 1$

$-\frac{1}{1} + \frac{1}{3} - \frac{1}{1}$

$-\frac{3}{3} + \frac{1}{3} - \frac{3}{3}$

$\boxed{-\frac{5}{3}}$

③ $-z^4 + 7(z+1)^2 - 5,\ z = 2$

$-2^4 + 7(2+1)^2 - 5$

$-16 + 7(3)^2 - 5$

$-16 + 7(9) - 5$

$-16 + 63 - 5$

$\boxed{42}$

④ $\frac{1}{x} + \frac{x^2+3}{7},\ x = 2$

$\frac{1}{2} + \frac{2^2+3}{7}$

$\frac{1}{2} + \frac{4+3}{7}$

$\frac{1}{2} + \frac{7}{7}$

$\frac{1}{2} + 1 = \boxed{\frac{3}{2}}$

⑤ $x^2 + 8 - x^3 + 6x,\ x = -1$

$(-1)^2 + 8 - (-1)^3 + 6(-1)$

$1 + 8 - (-1) - 6$

$1 + 8 + 1 - 6$

$\boxed{4}$

⑥ $-y^4 + \frac{3}{y+1},\ y = -2$

$-(-2)^4 + \frac{3}{-2+1}$

$-16 + \frac{3}{-1}$

$-16 - 3$

$\boxed{-19}$

⑦ $z^2 - 7z + 2z^3 + 4,\ z = 3$

$3^2 - 7(3) + 2(3)^3 + 4$

$9 - 21 + 2(27) + 4$

$9 - 21 + 54 + 4$

$\boxed{46}$

⑧ $-x^4 + 3x^2 - 1,\ x = 2$

$-2^4 + 3(2)^2 - 1$

$-16 + 3(4) - 1$

$-16 + 12 - 1$

$\boxed{-5}$

⑨ $3y^2 - y + \frac{5}{2y},\ y = -1$

$3(-1)^2 - (-1) + \frac{5}{2(-1)}$

$3(1) + 1 + \left(\frac{5}{-2}\right)$

$3 + 1 - \frac{5}{2}$

$\boxed{\frac{3}{2}}$

⑩ $-z^3 - z^2 + 10,\ z = -2$

$-(-2)^3 - (-2)^2 + 10$

$-(-8) - 4 + 10$

$8 - 4 + 10$

$\boxed{14}$

⑪ $-5(1-z) - 2z^3,\ z = -3$

$-5(1-(-3)) - 2(-3)^3$

$-5(1+3) - 2(-3)^3$

$-5(4) - 2(-27)$

$-20 + 54$

$\boxed{34}$

⑫ $-5 - 10x^2 - x - 3x^3,\ x = -2$

$-5 - 10(-2)^2 - (-2) - 3(-2)^3$

$-5 - 10(4) + 2 - 3(-8)$

$-5 - 40 + 2 + 24$

$\boxed{-19}$

13) $-3x^2 + \frac{-4}{3x-1} - x,\ x=-3$

$-3(-3)^2 + \frac{-4}{3(-3)-1} - (-3)$

$-3(9) + \frac{-4}{-9-1} + 3$

$-27 + \frac{-4}{-10} + 3$

$-\frac{27}{1} + \frac{2}{5} + \frac{3}{1} = -\frac{135}{5} + \frac{2}{5} + \frac{15}{5} = \boxed{\frac{-118}{5}}$

14) $7(y-y^2) - y - 2y^4,\ y=-1$

$7(-1-(-1)^2) - (-1) - 2(-1)^4$

$7(-1-1) + 1 - 2(1)$

$7(-2) + 1 - 2$

$-14 + 1 - 2 = \boxed{-15}$

15) $-x + x(5x + x^4) - 3x,\ x=-1$

$-(-1) - 1(5(-1) + (-1)^4) - 3(-1)$

$1 - 1(-5+1) + 3$

$1 - 1(-4) + 3$

$1 + 4 + 3$

$\boxed{8}$

16) $\frac{-y^2 - 3y}{7} - y,\ y=-4$

$\frac{-(-4)^2 - 3(-4)}{7} - (-4)$

$\frac{-16+12}{7} + 4$

$-\frac{4}{7} + \frac{4}{1} = \boxed{\frac{24}{7}}$

17) $4 + xy,\ x=-5, y=-3$

$4 + (-5)(-3)$

$4 + 15$

$\boxed{19}$

18) $5x + 2y,\ x=1, y=-6$

$5(1) + 2(-6)$

$5 - 12$

$\boxed{-7}$

19) $(x^2y - 4xy)^2 - 5x,\ x=2, y=-1$

$(2^2(-1) - 4(2)(-1))^2 - 5(2)$

$(4(-1) - 8(-1))^2 - 10$

$(-4+8)^2 - 10$

$4^2 - 10$

$16 - 10$

$\boxed{6}$

20) $x^2 + y^2,\ x=2, y=4$

$2^2 + 4^2$

$4 + 16$

$\boxed{20}$

Evaluating Real World Formulas

21) Find E below if $m=10$ and $c=4$.

$E = mc^2 = 10(4)^2 = 10(16) = \boxed{160}$

22) Find V below if $I=2.1$ and $r=0.8$.

$V = Ir = 2.1(0.8) = \boxed{1.68}$

23) Find P below if $W=6$ and $L=8$.

$P = 2L + 2W = 2(8) + 2(6) = 16 + 12 = \boxed{28}$

24) Find A below if $b=10$ and $h=20$.

$A = \frac{bh}{2} = \frac{10(20)}{2} = \frac{200}{2} = \boxed{100}$

(25) Find r below if $d=36.45$ and $t=8.1$.

$$r=\frac{d}{t}=\frac{36.45}{8.1}=\boxed{4.5}$$

(26) Find P below if $\pi=3.14$ and $r=9$.

$$P=2\pi r=2(3.14)(9)=18(3.14)=\boxed{56.52}$$

(27) Find V_x below if $b=6$ and $a=-2$.

$$V_x=-\frac{b}{2a}=-\frac{6}{2(-2)}=-\frac{6}{-4}=\frac{6}{4}=\boxed{\frac{3}{2}}$$

(28) Find D below if $a=2$, $b=-3$, and $c=-4$.

$$D=b^2-4ac=(-3)^2-4(2)(-4)=9-8(-4)=9+32$$
$$=\boxed{41}$$

(29) Find F below if $m=8$ and $a=-6$.

$$F=ma=8(-6)=\boxed{-48}$$

(30) Find K below if $m=6$ and $v=1$.

$$k=\frac{mv^2}{2}=\frac{6(1)^2}{2}=\frac{6(1)}{2}=\frac{6}{2}=\boxed{3}$$

(31) Find V below if $L=3$, $W=2$, and $H=3$.

$$V=LWH=3(2)(3)=6(3)=\boxed{18}$$

(32) Find V below if $V_0=-3$, $a=4.5$, and $t=18$.

$$V=V_0+at=-3+4.5(18)=-3+81=\boxed{78}$$

(33) Find A below if $P=812$, $r=0.09$, and $t=4$.

$$A=P+Prt=812+812(0.09)(4)=812+292.32=1{,}104.32$$

(34) Find d below if $V_0=-2$, $a=-3$, $t=5$, and $x_0=8$.

$$d=x_0+V_0t+\frac{at^2}{2}=8-2(5)+\frac{-3(5)^2}{2}=8-10+\frac{-3(25)}{2}=\frac{-2}{1}-\frac{75}{2}=\boxed{\frac{-79}{2}}$$

(35) Find F below if $G=2$, $M_1=2$, $M_2=3$, and $d=5$.

$$F=\frac{GM_1M_2}{d^2}=\frac{2(2)(3)}{5^2}=\boxed{\frac{12}{25}}$$

(36) Find C below if $F=20$.

$$C=\frac{5}{9}(F-32)=\frac{5}{9}(20-32)=\frac{5}{9}(-12)=\frac{5}{9}\left(\frac{-12}{1}\right)=\frac{-60}{9}=\boxed{\frac{-20}{3}}$$

(37) Find A below if $b_1=3$, $b_2=5$, and $h=1$.

$$A=\frac{h(b_1+b_2)}{2}=\frac{1(3+5)}{2}=\frac{1(8)}{2}=\frac{8}{2}=\boxed{4}$$

(38) Find A below if $P=722$, $r=0.05$, $n=2$, and $t=2$. Round your answer to the nearest hundredth.

$$A=P\left(1+\frac{r}{n}\right)^{nt}=722\left(1+\frac{0.05}{2}\right)^{2(2)}$$
$$=722(1+0.025)^4$$
$$=722(1.025)^4$$
$$=722(1.10381289)$$
$$\approx\boxed{796.953}$$

$$\boxed{X^Y}$$

Checking Solutions

(39) $3x-1=-4x+34,\ x=5$

$3(5)-1=-4(5)+34$

$15-1=-20+34$

$\boxed{14=14}$

$\boxed{\text{Yes}}$

(40) $-\frac{y}{2}+4y-1=7,\ y=-2$

$-\left(\frac{-2}{2}\right)+4(-2)-1=7$

$-(-1)-8-1=7$

$1-8-1=7$

$\boxed{-8\neq 7}$

$\boxed{\text{No}}$

(41) $\frac{z+2}{3}-1=3z-11,\ z=4$

$\frac{4+2}{3}-1=3(4)-11$

$\frac{6}{3}-1=12-11$

$2-1=12-11$

$\boxed{1=1}$ $\boxed{\text{Yes}}$

(42) $x^2-2x+1=2,\ x=2$

$2^2-2(2)+1=2$

$4-2(2)+1=2$

$4-4+1=2$

$\boxed{1\neq 2}$

$\boxed{\text{No}}$

(43) $y-9=3y+1,\ y=-5$

$-5-9=3(-5)+1$

$-14=-15+1$

$\boxed{-14=-14}$

$\boxed{\text{Yes}}$

(44) $6+2x+(-3)=-x+5x-10,\ x=-1$

$6+2(-1)+(-3)=-(-1)+5(-1)-10$

$6-2-3=1-5-10$

$\boxed{1\neq -14}$

$\boxed{\text{No}}$

(45) $\frac{2z}{3}-7=4z,\ z=2$

$\frac{2(2)}{3}-7=4(2)$

$\frac{4}{3}-\frac{7}{1}=8$

$\frac{4}{3}-\frac{21}{3}=8$

$\boxed{-\frac{17}{3}\neq 8}$ $\boxed{\text{NO}}$

(46) $-x+1=3x-10x+7,\ x=1$

$-1+1=3(1)-10(1)+7$

$0=3-10+7$

$0=-7+7$

$\boxed{0=0}$

$\boxed{\text{Yes}}$

(47) $2y-5y+4=7y+2,\ y=0$

$2(0)-5(0)+4=7(0)+2$

$0-0+4=0+2$

$0+4=2$

$\boxed{4\neq 2}$

$\boxed{\text{NO}}$

(48) $-2z+30=-8z-4z,\ z=-3$

$-2(-3)+30=-8(-3)-4(-3)$

$6+30=24+12$

$\boxed{36=36}$

$\boxed{\text{Yes}}$

(49) $x-36+3x=-3x-2x,\ x=-3$

$-3-36+3(-3)=-3(-3)-2(-3)$

$-39-9=9+6$

$\boxed{-48\neq 15}$

$\boxed{\text{NO}}$

(50) $-y^2+1=7y+2,\ y=-3$

$-(-3)^2+1=7(-3)+2$

$-9+1=-21+2$

$\boxed{-8\neq -19}$

$\boxed{\text{NO}}$